

AI-Powered Enterprise Journey Generation Platform

SUMMARY

The platform converts a simple user request into a validated, production-ready digital journey by combining enterprise knowledge, AI reasoning, reusable components, design systems, APIs, compliance rules, and continuous learning.

USER REQUEST & CHANNEL LAYER

The journey can start from any supported channel or platform: PayZapp, MyCards, Mobile Banking / Net Banking, Adobe, NovoPay, Future Platforms, API Calls, Chat Interfaces. Example user request: "I want to change my address." The request enters the Intent Understanding Layer.

INTENT UNDERSTANDING

Intent Understanding Agent identifies: Intent, Platform, Journey Type, Entities, Confidence, Priority. Example AI understanding for "I want to change my address.": Intent UPDATE_ADDRESS, Platform MyCards, Journey Type Profile Update, Confidence 99%. Confidence above threshold: yes continue to journey generation, no send to Unknown Intent Handler.

UNKNOWN INTENT HANDLER

If the AI does not confidently understand the request, the system does not guess. Discovery sequence: Internal Knowledge Registry → Enterprise Knowledge → Official HDFC Repositories → Public Product Pages → FAQs → Documentation. If not found: Respond Gracefully → Create Knowledge Request → Add to Learning / Knowledge Queue. Key principle: If the system does not know, it does not hallucinate.

CACHED KNOWLEDGE PACK

Before new retrieval, the platform checks whether the same journey has already been solved recently. Cache key: Intent + Platform + Journey Version (example: UPDATE_ADDRESS + MyCards + V3). Cache hit: reuse existing Knowledge Pack. Cache miss: start fresh retrieval. Improves response latency, token efficiency, LLM cost, consistency, and reusability.

ENTERPRISE KNOWLEDGE REGISTRY

The Knowledge Registry is the index of the enterprise experience ecosystem and stores references and metadata rather than full documents. Example mapping for UPDATE_ADDRESS links Journey_023, Validation_19, components Input_Address/OTP/Confirmation, API Address_API, Compliance_2.3, Content_v4, Tracking_Spec, MyCards_Theme.

KNOWLEDGE RETRIEVAL ENGINE

Retrieves only knowledge required for the identified journey from sources such as journey documentation, markdown/technical documentation, design system, component library, API specifications, validation rules, accessibility guidelines, compliance rules, brand guidelines, content guidelines, analytics specs, error libraries, and navigation rules. Output is a Cached Knowledge Pack.

JOURNEY PLANNER AGENT

Reasoning layer that converts enterprise knowledge into a structured journey plan. Determines screens, sequence, navigation, dependencies, required fields, validation, business rules, API calls, error handling, success flow, analytics and accessibility requirements. Produces a Journey Blueprint and does not directly create UI.

RESPONSE ORCHESTRATOR

Defines user experience and conversational behavior: what EVA says, screen content, button labels, CTAs, loading states, error messages, snackbars, empty states, success messages, voice responses, localized content. Produces experience response specification.

COMPONENT INTELLIGENCE ENGINE

Decision flow: Search Component Registry → if found, reuse existing component. If not found, search for similar components → if similar found, create variant/compose existing components; else generate new component. Core principle: Generation is last option, reuse is first option.

DESIGN SYSTEM ENGINE

Ensures generated experience complies with enterprise design system using design tokens, colours, typography, spacing, grid, icons, illustrations, motion, themes, accessibility, platform variants, and brand rules.

DETERMINISTIC JSON COMPILER

Inputs: Journey Blueprint + Response Specification + Selected Components. Deterministically produces frontend JSON, EVA JSON, navigation, actions, API bindings, analytics events. Principle: not an AI reasoning layer; transforms plan into executable experience definition for reliability, consistency, and testability.

AI VALIDATION ENGINE

Automated validation before production across Experience (navigation, responsive behaviour, component usage, spacing, naming), Design (brand compliance, design system compliance, contrast, accessibility), Technical (JSON validity, API bindings, broken references, missing assets, performance), Enterprise (compliance, platform rules, analytics requirements). PASS continues to output; FAIL returns for correction and revalidation. Principle: nothing moves forward until required validations pass.

OUTPUT ENGINE

After validation, generates implementation formats based on available connectors: Figma, Adobe XD, Flutter, React, Angular, Android XML, iOS SwiftUI, Web Components, Frontend JSON.

HUMAN-IN-THE-LOOP

AI-generated experiences are reviewed by designer, product manager, developer, and compliance team. Approval flow: Generated Experience → Human Review → Approved: Publish; Not approved: Capture feedback, send to Learning Queue, improve future recommendations.

LEARNING ENGINE

Every decision creates reusable intelligence. Approved experiences teach what to reuse; rejected experiences teach what to avoid. Metadata tracked includes popularity, reuse count, approval rate, performance, accessibility score, owner, last updated. Over time improves right journey/component selection without continuous model retraining.

KNOWLEDGE INGESTION PIPELINE

Continuously incorporates new enterprise knowledge for new journey, documentation, component, API, theme, guardrail, compliance rule. Pipeline performs: Chunk → Tag → Version → Embed → Index → Update Knowledge Registry → Refresh Cache.

OBSERVABILITY & DESIGNOPS

Tracks performance (journey generation latency, generation time, API performance, validation time), AI efficiency (LLM calls, token usage, cache hit %, cost per journey), reusability (component reuse %, new component creation %, most used components), quality (validation failure rate, approval rate, designer edits, accessibility score), and knowledge (unknown intents, most requested features, knowledge gaps).

COMPLETE END-TO-END FLOW

Flow from user/channel through intent understanding, unknown intent handling (when low confidence), cache check, knowledge retrieval and pack, journey planner and blueprint, response orchestrator and component intelligence, design system engine, JSON compiler, AI validation engine (pass/fail and revalidate), output engine, human-in-the-loop, publish or learning queue, learning engine, improved future reuse.

CORE VALUE PROPOSITION

Controlled loop: Understand → Retrieve → Plan → Reuse → Generate → Validate → Approve → Publish → Learn. Combines enterprise knowledge + reusable assets + AI reasoning + deterministic compilation + automated validation + human governance. Results: faster journey creation, higher reuse, lower AI cost, consistent design, enterprise compliance, reduced hallucination, multi-platform delivery, continuous improvement.